

Project Outline and Core Findings

The project in one paragraph

We are developing a 2-DOF finger exoskeleton that sits around the hand and actuates the thumb and index finger. Surface EMG (muscle) sensors detect Parkinsonian tremor, and when shaking is detected during grasping, the exoskeleton stiffens the fingers to damp the tremor while leaving slow, voluntary motion free. The concept exploits two facts from the literature: tremor (4-6 Hz) and voluntary motion (below about 2 Hz) are separable in frequency, and muscle activity precedes mechanical shaking by tens of milliseconds - so an EMG-triggered device can stiffen before the oscillation grows.

The problem: Parkinsonian tremor and grasping

- Parkinson's disease affected about 11.8 million people worldwide in 2021, projected to reach about 25 million by 2050. Rest tremor affects roughly 65-75% of patients.
- Classic Parkinsonian rest tremor is 4-6 Hz pill-rolling of thumb and index finger - largest at the fingers, exactly the digits we target.
- Crucially for grasping: re-emergent tremor means rest tremor reappears about 10 seconds after adopting a holding posture (cup, spoon), at the same frequency. Holding objects is therefore not protected by the usual suppression of tremor during movement.
- Impact: in a 2025 survey (634 respondents), 83% said tremor affects at least one daily activity (handwriting 60%, keyboard 52%, smartphone 47%); 92% reported psychosocial impact.
- Treatments leave a gap: about 39% of patients have tremor resistant to levodopa; deep brain stimulation helps 70-80% but is invasive, costs around 40,000 USD, and only a minority qualify; focused ultrasound is unilateral and irreversible. This treatment-resistant group is the target population for assistive devices.

What exists today (state of the art in brief)

- Commercial tremor devices act at the wrist or on the utensil, not the fingers: GyroGear GyroGlove (gyroscopic, 5,899 USD), Steadiwear (tuned damper glove), Liftware (self-stabilizing spoon, 71-76% reduction), Cala kIQ (wrist nerve stimulation, about 64% median improvement in PD action tremor). None stiffen individual fingers, and several have no published clinical trials.
- Research orthoses: WOTAS (the canonical robotic tremor orthosis for elbow, forearm and wrist) achieved about 40% average and up to 80% tremor power reduction; across 25+ devices, reviews report roughly 70-86% suppression in human tests. Notably, 92% of devices sense motion (IMU) only - only one used EMG.
- Hand exoskeletons are a mature field, but for stroke rehabilitation and grip assistance (e.g. Bioservo Carbonhand, Harvard soft glove, Hand of Hope). Thumb + index is a recognized minimal grasp set. Standard specs: 1-2 DOF per finger, under 300-500 g on the hand, joint torques of 0.1-1.4 Nm.

- Closest prior art: the Wearable Tremor Suppression Glove (Western University, 2021), motor-driven suppression at thumb/index MCP and wrist with about 85% amplitude reduction, and a 30 g layer-jamming stiffening glove (25-42% clinical reduction, always on). Neither is EMG-triggered nor grasp-contingent.

The gap we fill

No existing device combines: (1) finger-level action (thumb + index), (2) stiffening (semi-active damping) rather than motorized counter-motion, and (3) EMG-based detection so stiffening happens only when tremor appears during grasping. EMG-in-the-loop mechanical tremor suppression is essentially unexplored - a defensible novelty for a university project.

Control strategy (options and recommendation)

- Strategy families: passive damping (always on - hurts voluntary motion), active cancellation (motors counter the shake; most effective but heavy, power-hungry, and able to inject energy), semi-active stiffening (modulates stiffness/damping only; cannot amplify tremor - inherently safe), and FES (electrical muscle stimulation; effective but fatiguing).
- Our choice, supported by the literature: semi-active impedance modulation - render extra stiffness/damping about the estimated voluntary motion when tremor is detected. The equivalent WOTAS mode achieved about 70% power reduction.
- Signal separation is standard: voluntary motion below 2 Hz, tremor 4-6 Hz. Real-time estimators (WFLC/BMFLC adaptive filters, Kalman two-stage schemes) track tremor with about 1 ms effective lag and run comfortably on a microcontroller.
- Safety: cap finger-joint torque near 0.1 Nm actively (hard stops at anatomical limits), semi-active elements only resist motion, and monitor that suppression does not push tremor to proximal joints (seen in 1 of 10 WOTAS patients).

Muscle sensing in brief (full detail in the Detailed Description doc)

- What we measure: surface EMG from forearm muscles driving the fingers - flexor digitorum superficialis and extensor digitorum communis for the index; flexor pollicis longus / thenar (abductor pollicis brevis) and first dorsal interosseous for the thumb side. All are surface-accessible; plan 4-8 channels rather than a single flexor/extensor pair.
- Pipeline: band-pass 20-450 Hz, mains interference handling, rectify, envelope (RMS window around 100-250 ms for intent; a faster 8-10 Hz envelope to see tremor bursts), normalize to a calibration contraction.
- Tremor signature: rhythmic EMG bursts at 4-6 Hz, alternating between antagonist muscles in PD (synchronous in essential tremor - a built-in differentiator). Detection = band-power threshold on the envelope spectrum, or burst-onset detectors (TKEO).
- The killer feature: EMG precedes mechanical tremor by roughly 20-100 ms (electromechanical delay), so the device can stiffen predictively, before the shake builds up.

- Grasp intent: co-activation of flexors above threshold means the user is grasping, which arms the stiffening. The same EMG-triggered logic is proven in commercial grip-assist gloves.

Open questions and risks

- Can 2 DOF (one per digit) provide enough coverage, given tremor also expresses at the wrist? Literature says finger-level suppression works, but the wrist may need a future extension.
- Distinguishing tremor bursts from fast voluntary activity in EMG on real patients (medication ON/OFF states change both tremor and baseline muscle tone).
- Comfort and weight budget: target under roughly 200-300 g on the hand; donning with tremulous hands must be one-handed and simple.
- Access to patients for testing and ethics approval; initially validate on healthy subjects with simulated tremor, which is standard practice in the field.

Evaluation plan (benchmarks from literature)

- Primary metric: tremor power reduction in the 3.5-7.5 Hz band (accelerometer/IMU on the finger), device on vs off. Benchmarks: 40% (WOTAS average) to 70-86% (best human-tested orthoses).
- Task-level tests: grasp-and-hold (cup, spoon), nine-hole peg test, spiral drawing; task completion time and spill/error rates.
- EMG detector quality: detection latency (target below the electromechanical delay), sensitivity and false-positive rate during voluntary motion.
- Transparency: interaction resistance felt during voluntary motion with the device passive.

Suggested presentation skeleton

- Slide 1 - Motivation: PD numbers and what tremor does to grasping (video/demo).
- Slide 2 - Why existing solutions do not cover it: the drug/DBS gap; devices act at the wrist or utensil, not the fingers.
- Slide 3 - State of the art: tremor devices and hand exoskeletons, and the gap we fill.
- Slide 4 - Our concept: 2-DOF thumb+index exoskeleton with EMG-triggered stiffening (system diagram).
- Slide 5 - Muscle signals: acquisition, processing, tremor signature (one signal figure).
- Slide 6 - Control strategy: frequency separation, semi-active stiffening, safety.
- Slide 7 - Evaluation plan and expected outcomes vs literature benchmarks.
- Slide 8 - Timeline, risks, and what we need (equipment, possible patient contact).